THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a leaf, the transport of food is blocked when a ring of bark is removed (girdling). The tissue blocked is: 1

(a) phloem **(b)** xylem **(c)** epidermis **(d)** cambium

2. Assertion (A): Phloem transport is an active process. Reason (R): Sucrose is loaded into phloem using energy from ATP. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. In anaerobic respiration in yeast, glucose is broken down into: 1

(a) water and carbon dioxide **(b)** ethanol and carbon dioxide
(c) lactic acid **(d)** pyruvate only

4. Which is correct about blood flow? Arteries carry blood: 1

(a) only deoxygenated blood **(b)** to the heart under low pressure
(c) only oxygenated blood **(d)** away from the heart under high pressure

5. Assertion (A): Stomata remain open in daytime in most plants. Reason (R): Guard cells lose water in the day. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The first step of breakdown of glucose, into pyruvate, occurs in the: 1

(a) mitochondria **(b)** nucleus **(c)** cytoplasm **(d)** chloroplast

7. Artificial kidney in dialysis uses a membrane to remove: 1

(a) urea **(b)** proteins **(c)** glucose **(d)** oxygen

8. Assertion (A): Dialysis is used in a patient with kidney failure. Reason (R): The machine removes urea and extra salts from the blood. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** The small intestine has a large surface area for absorption. 1
Reason (R): The inner lining of the small intestine has numerous finger-like projections called villi.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Filtration of blood in the kidney takes place in the glomerulus. 1
Reason (R): The ureter carries urine from the kidney to the urinary bladder.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the balanced equation for fermentation of glucose by yeast. 2

12. Why does blood from the lungs return to the left atrium through the pulmonary vein and what is unusual about it? 2

13. Why does the respiration of the diver in deep water need compressed air? 2

14. Differentiate between the pulmonary and systemic circulation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If the diaphragm of a person is paralysed, which phase of breathing is affected and why? 3

16. Explain how the level of oxygen in the blood is regulated during exercise. 3

17. Explain the role of the diaphragm and rib muscles in changing lung volume. 3

OR

Describe how alveoli and haemoglobin together help in oxygen transport. 3

18. Why is it dangerous if a person's urine output stops completely? 3

OR

Explain why a mammal needs a double circulation while a fish with a single circulation survives. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mr Singh's kidneys stopped working properly, so he visits a hospital every week for haemodialysis. The machine contains a tube lined with a semi-permeable membrane and surrounded by a dialysing fluid. The fluid has the same osmotic pressure as blood but contains no nitrogenous waste. 5
(i) Name the structural and functional unit of the kidney.
(ii) Name the tube that carries urine from the kidney to the urinary bladder.
(iii) Name the nitrogenous waste mainly removed from the blood.
(iv) Why does the dialysing fluid contain no nitrogenous waste and why is its osmotic pressure the same as blood?

20. Explain the digestion of fats from the mouth to the absorption, including bile, lipase, emulsification and lymph. 5

OR

A patient's kidneys are failing. Explain the effects on the body and how dialysis helps, including its limitation. 5

END OF QUESTION PAPER — Total: 40 Marks